

Outline

- Probability theory - basics

- Probabilistic Inference

 - Inference by enumeration

- Conditional Independence

- Bayesian Networks

- Inference in Bayesian Networks

 - Exact Inference

 - Approximate Inference

Probabilistic Inference in BNs

- ❑ Graphical independence representation yields efficient inference schemes
- ❑ We generally want to compute
 - $P(X|E)$, where E is the evidence from the sensory measurements (known values for variables)
 - Sometimes, we may want to compute just $P(X)$
- ❑ Three approaches
 - Enumeration
 - Variable Elimination
 - Approximate Inference
 - sampling

Inference by Enumeration

- General case - the set of random variables $X_1, \dots, X_n$ is broken into three categories
 - Evidence variables - $E_1, \dots, E_k = e_1, \dots, e_k$
 - Query variables - $Q_1, \dots, Q_L = Q$
 - Hidden variables - $H_1, \dots, H_r$
- We want to estimate - $P(Q|e_1, \dots, e_k)$

Inference by Enumeration

- ❑ We want to estimate -
 $P(Q|e_1, \dots, e_k)$
- ❑ Step 1 – select the consistent entries with the evidence
- ❑ Step 2 - sum of the hidden variables to get the joint of Query and evidence

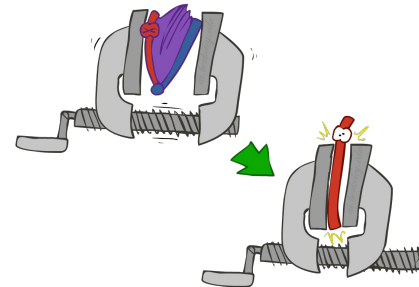
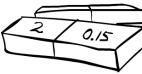
- $P(Q, e_1, \dots, e_k) = \sum_{H_1, \dots, H_r} P(Q, h_1, \dots, h_r, e_1, \dots, e_k)$

- ❑ Step 3 – Normalize

- $Z = \sum_q P(Q, e_1, \dots, e_k)$
 - $P(Q|e_1, \dots, e_k) = \frac{1}{Z} P(Q, e_1, \dots, e_k)$



x	P(x)
-3	0.05
-1	0.25
0	0.07
1	0.2
5	0.01



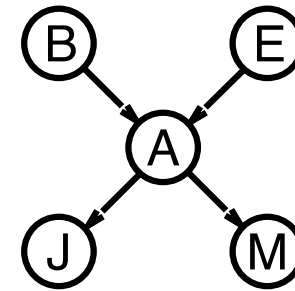
$$\times \frac{1}{Z}$$

Inference by Enumeration

- Slightly intelligent way to sum out variables from the joint without actually constructing its explicit representation.

- Consider the simple query on the burglary network

- $P(B|j, m)$



Inference by Enumeration (2)

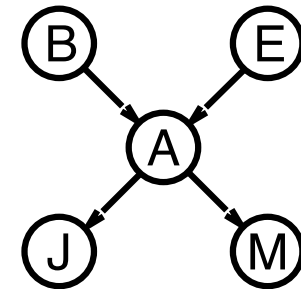
□ Slightly intelligent way to sum out variables from the joint without actually constructing its explicit representation.

□ Consider the simple query on the burglary network

○ $P(B|j, m) = P(B, j, m) / P(j, m)$

○ $= \frac{1}{Z} P(B, j, m)$

○ $= \frac{1}{Z} \sum_e \sum_a P(B, e, a, j, m)$



□ Rewrite full joint entries using product of the CPT entries

Inference by Enumeration (3)

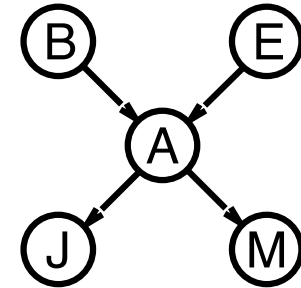
□ Slightly intelligent way to sum out variables from the joint without actually constructing its explicit representation.

□ Consider the simple query on the burglary network

$$\circ P(B|j, m) = P(B, j, m) / P(j, m)$$

$$\circ = \frac{1}{Z} P(B, j, m)$$

$$\circ = \frac{1}{Z} \sum_e \sum_a P(B, e, a, j, m)$$



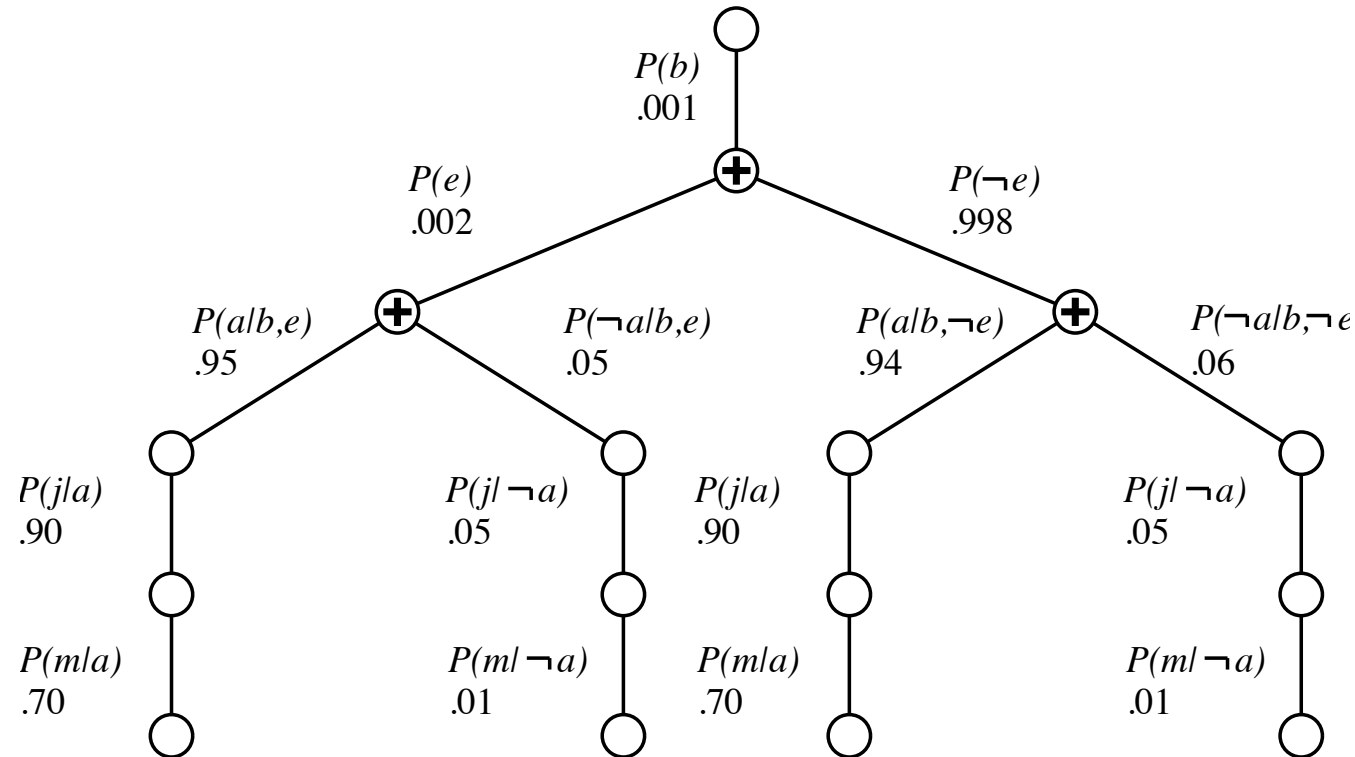
□ Rewrite full joint entries using product of the CPT entries

$$\circ = \frac{1}{Z} P(B) \sum_e P(e) \sum_a P(a|B, e) P(j|a) P(m|a)$$

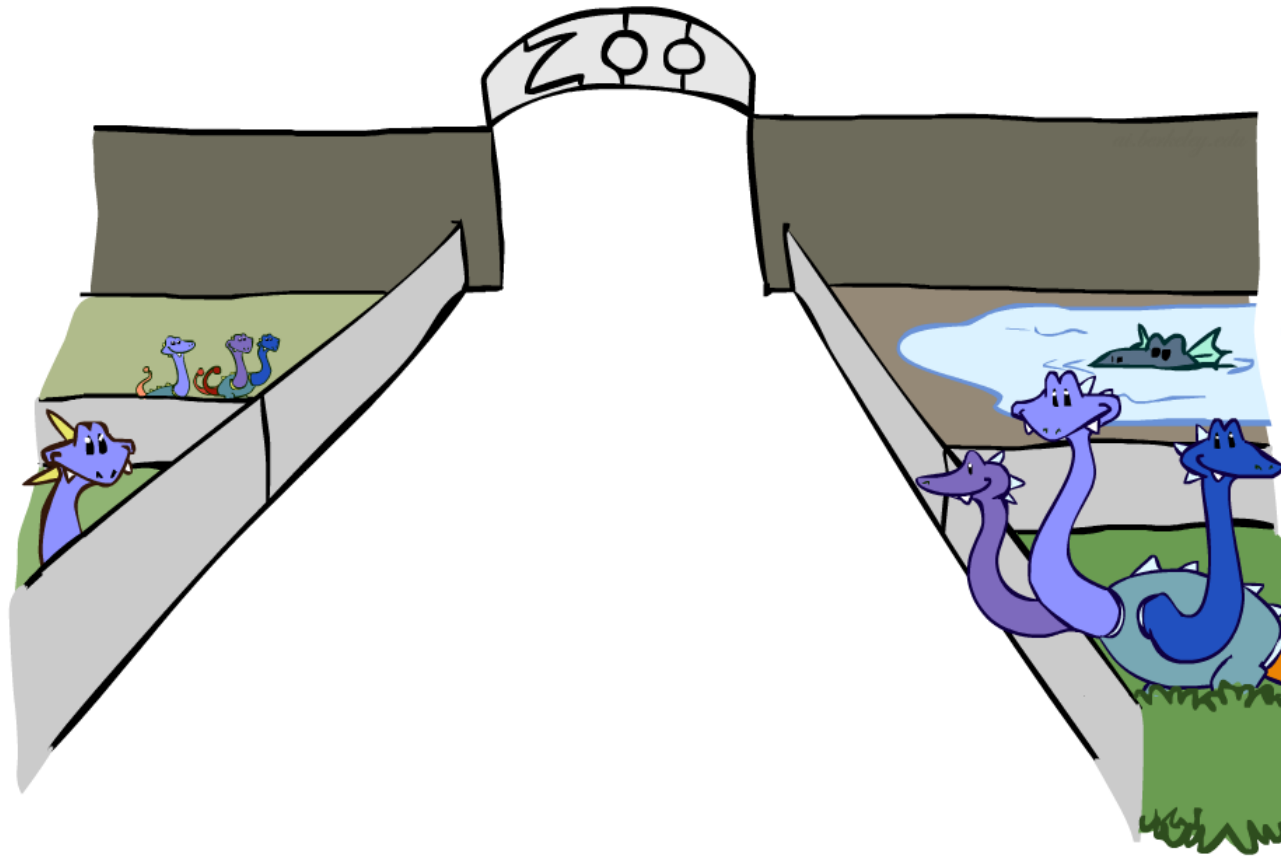
Evaluation Tree

Recursive depth first enumeration

$$\bigcirc = \frac{1}{Z} P(B) \sum_e P(e) \sum_a P(a|B, e) P(j|a) P(m|a)$$



Factor Zoo



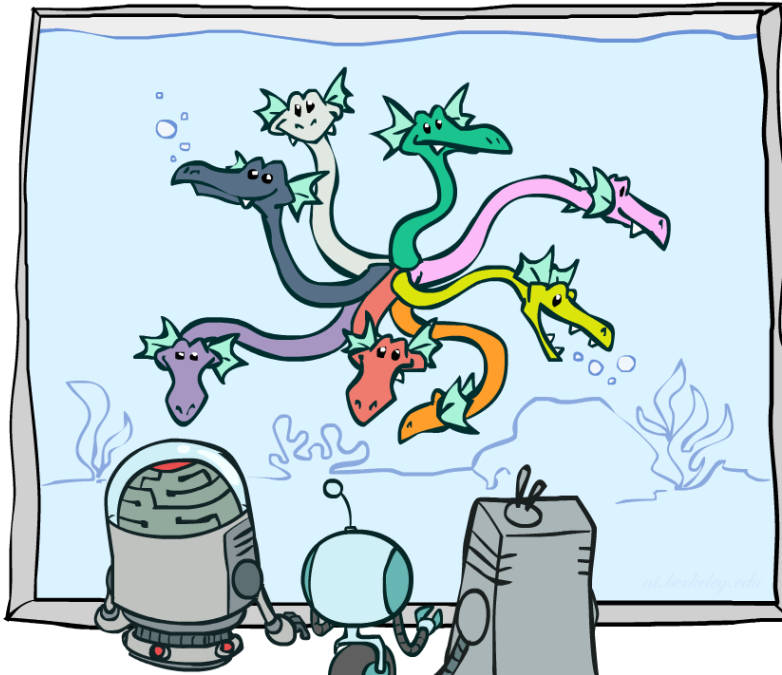
Factor Zoo I

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$P(\text{cold}, W)$

T	W	P
cold	sun	0.2
cold	rain	0.3



❑ Joint distribution: $P(X, Y)$

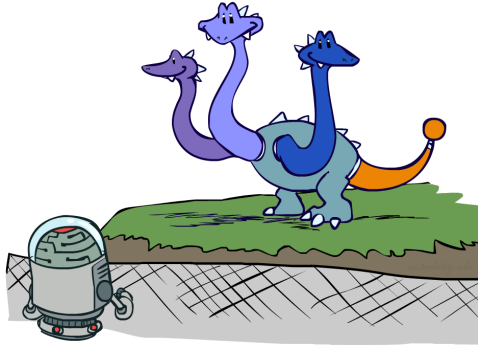
- Entries $P(x, y)$ for all x, y
- Sums to 1

❑ Selected joint: $P(x, Y)$

- A slice of the joint distribution
- Entries $P(x, y)$ for fixed x , all y
- Sums to $P(x)$

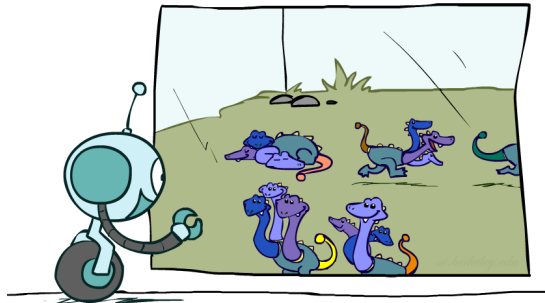
❑ Number of capitals determines dimensionality of the table

Factor Zoo II



$P(W|cold)$

T	W	P
cold	sun	0.4
cold	rain	0.6



$P(W|T)$

T	W	P
hot	sun	0.8
hot	rain	0.2
cold	sun	0.4
cold	rain	0.6

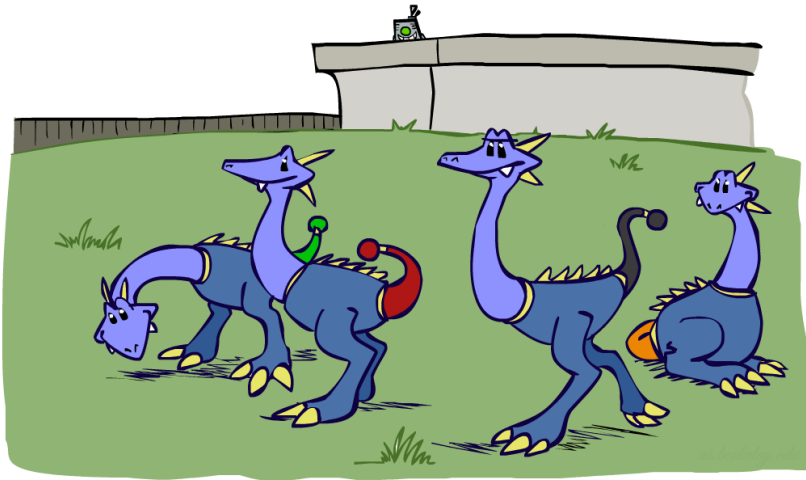
Single conditional: $P(Y | x)$

- Entries $P(y | x)$ for fixed x , all y
- Sums to 1

Family of conditionals: $P(Y | X)$

- Multiple conditionals
- Entries $P(y | x)$ for all x, y
- Sums to $|X|$

Factor Zoo III



$$P(\text{rain}|T)$$

T	W	P
hot	rain	0.2
cold	rain	0.6

❑ Specified family: $P(y | X)$

- Entries $P(y | x)$ for fixed y , but for all x
- Sums to ... who knows!

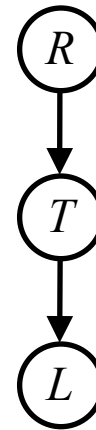
Example: Traffic Domain

Random Variables

- R : Raining
- T : Traffic
- L : Late for flight!

$P(L) =$

- $= \sum_{r,t} P(r, t, L)$
- $= \sum_{r,t} P(r)P(t|r)P(L|t)$



$P(R)$

+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Inference by Enumeration – Procedural Outline

- ❑ Procedure tracks objects called factors
- ❑ Initial factors are local CPTs (one per node)

$P(R)$		$P(T R)$			$P(L T)$		
+r	0.1	+r	+t	0.8	+t	+l	0.3
+r		+r	-t	0.2	+t	-l	0.7
-r	0.9	-r	+t	0.1	-t	+l	0.1
		-r	-t	0.9	-t	-l	0.9

- ❑ Any known values are selected

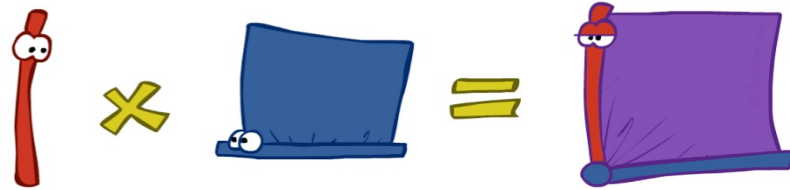
- For example if we know $L = \text{true}$,

$P(R)$		$P(T R)$			$P(+l T)$		
+r	0.1	+r	+t	0.8	+t	+l	0.3
+r		+r	-t	0.2	+t	-l	
-r	0.9	-r	+t	0.1	-t	+l	0.1
		-r	-t	0.9			

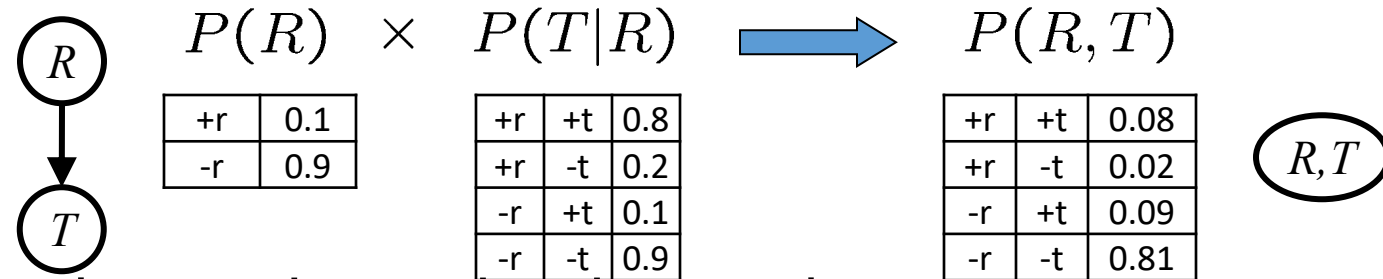
- ❑ Procedure – Join all factors and eliminate hidden variables

Operation 1 – Join Factors

- ❑ Similar to database join
- ❑ Get all factors over the joining variable
 - Build a new factor over the union of the variables involved

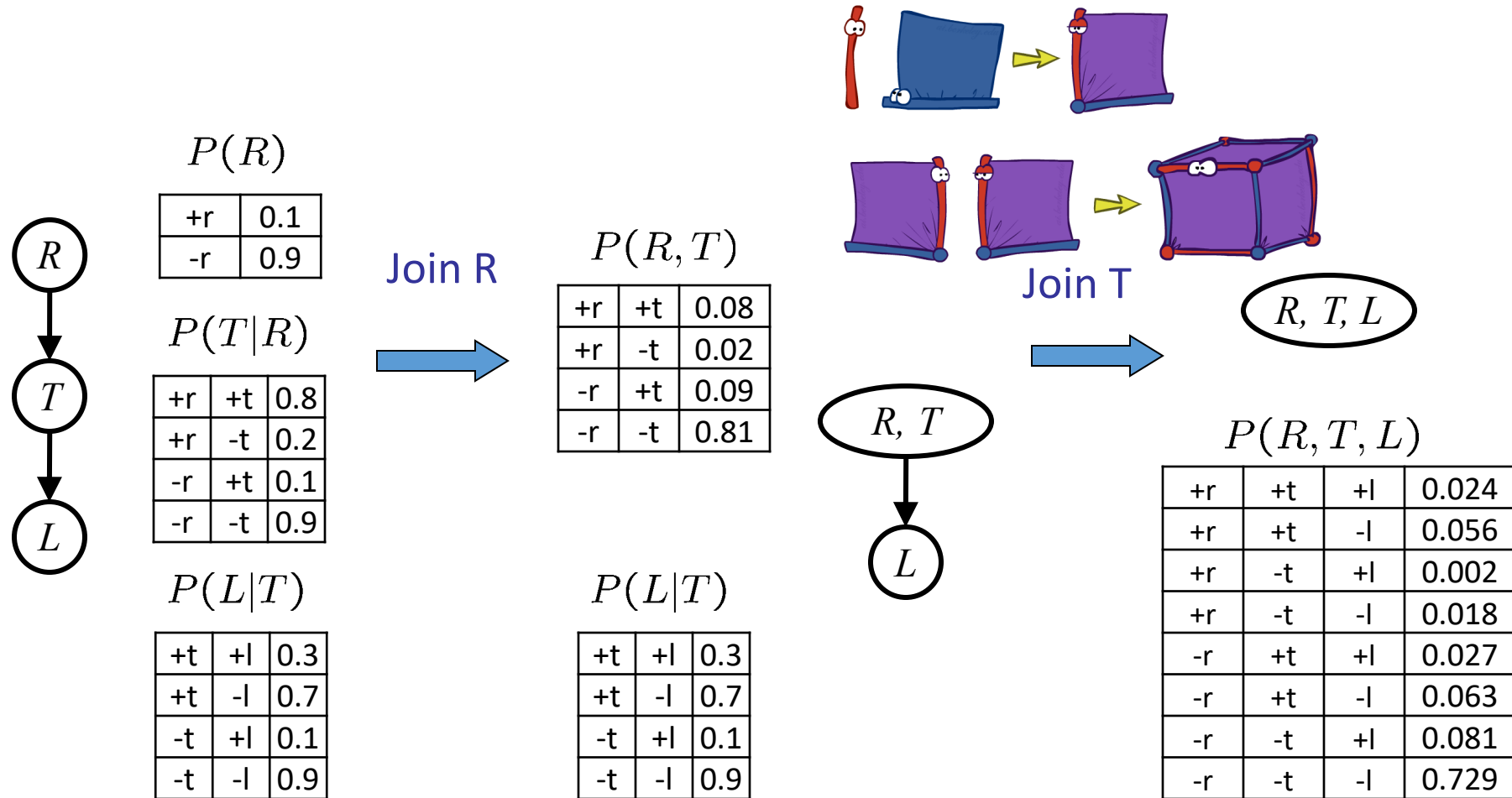


❑ Example – Join on R



- Computation of each entry is a pointwise product
 - $\forall r, t \ P(r, t) = P(r) \cdot P(t|r)$

Example: Multiple Joins



Operation 2: Eliminate

❑ Second basic operation: marginalization

❑ Take a factor and sum out a variable

- Shrinks a factor to a smaller one
- A projection operation

❑ Example:

$P(R, T)$

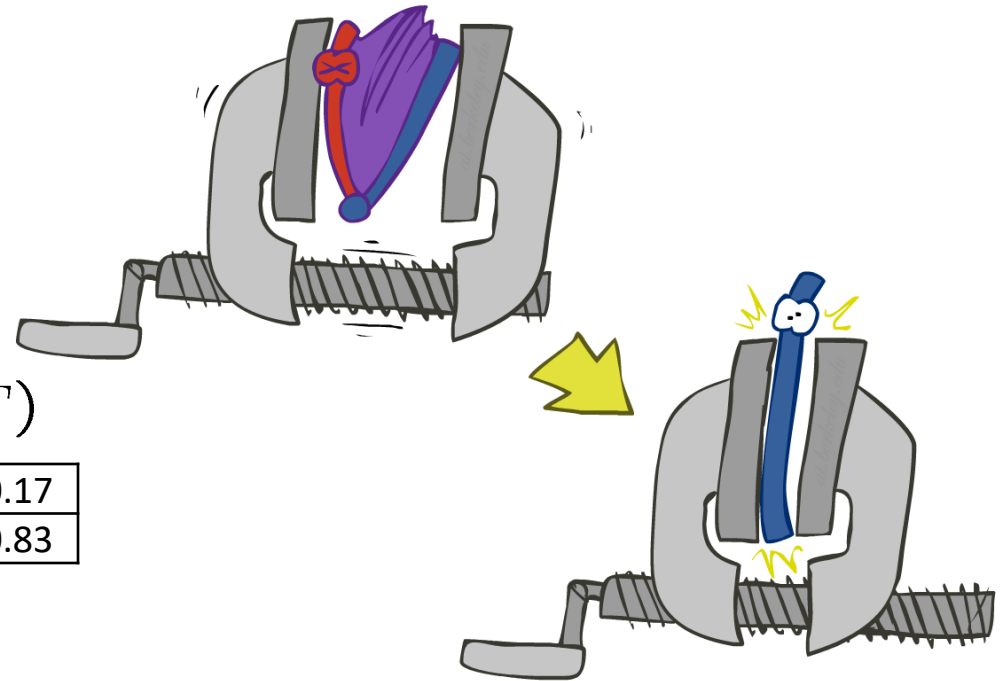
+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81

sum R



$P(T)$

+t	0.17
-t	0.83



Multiple Elimination

$P(R, T, L)$

R, T, L			
+r	+t	+l	0.024
+r	+t	-l	0.056
+r	-t	+l	0.002
+r	-t	-l	0.018
-r	+t	+l	0.027
-r	+t	-l	0.063
-r	-t	+l	0.081
-r	-t	-l	0.729

Sum
out R

→

$P(T, L)$

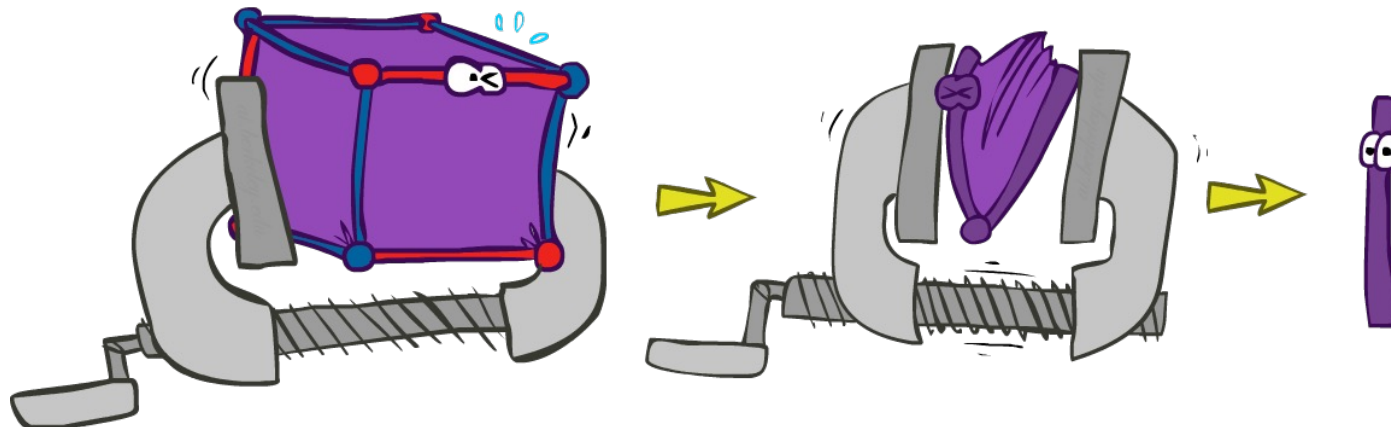
T, L		
+t	+l	0.051
+t	-l	0.119
-t	+l	0.083
-t	-l	0.747

Sum
out T

→

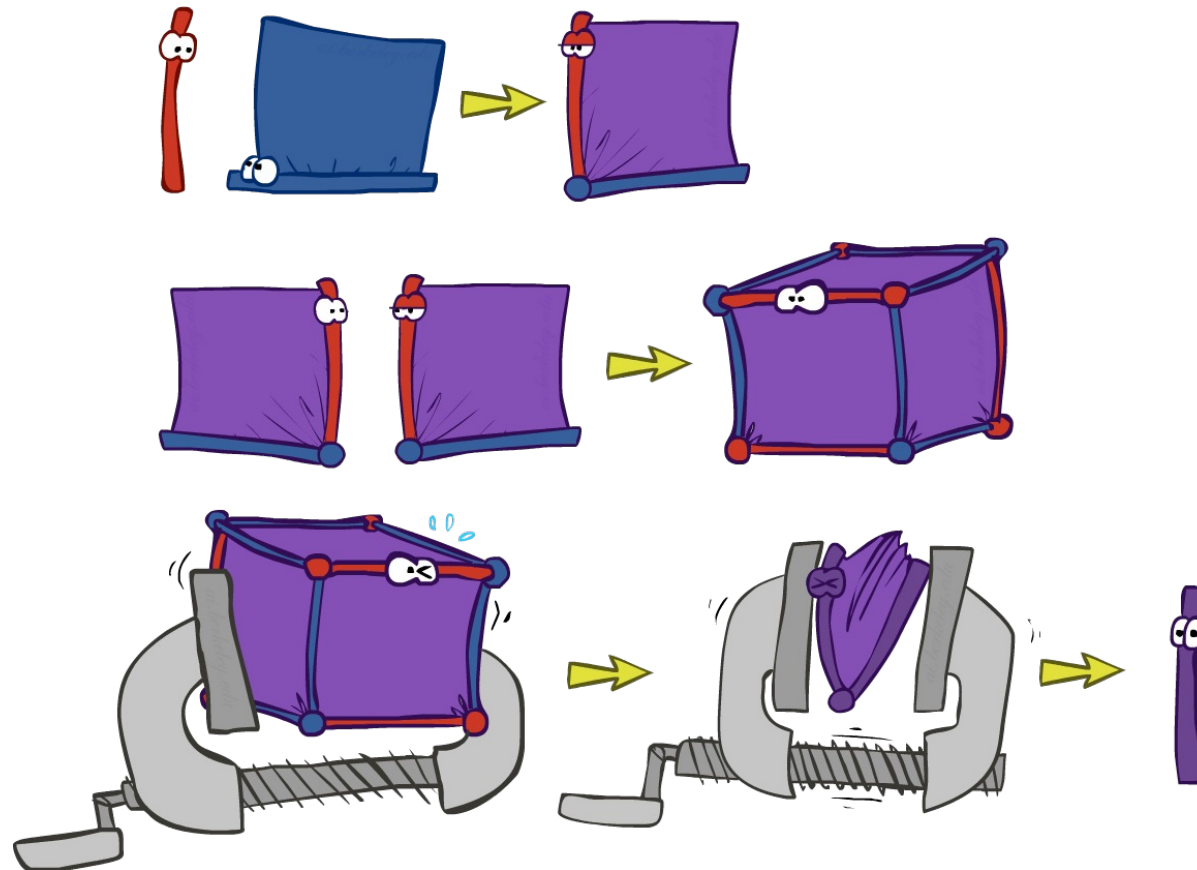
$P(L)$

L	
+l	0.134
-l	0.866

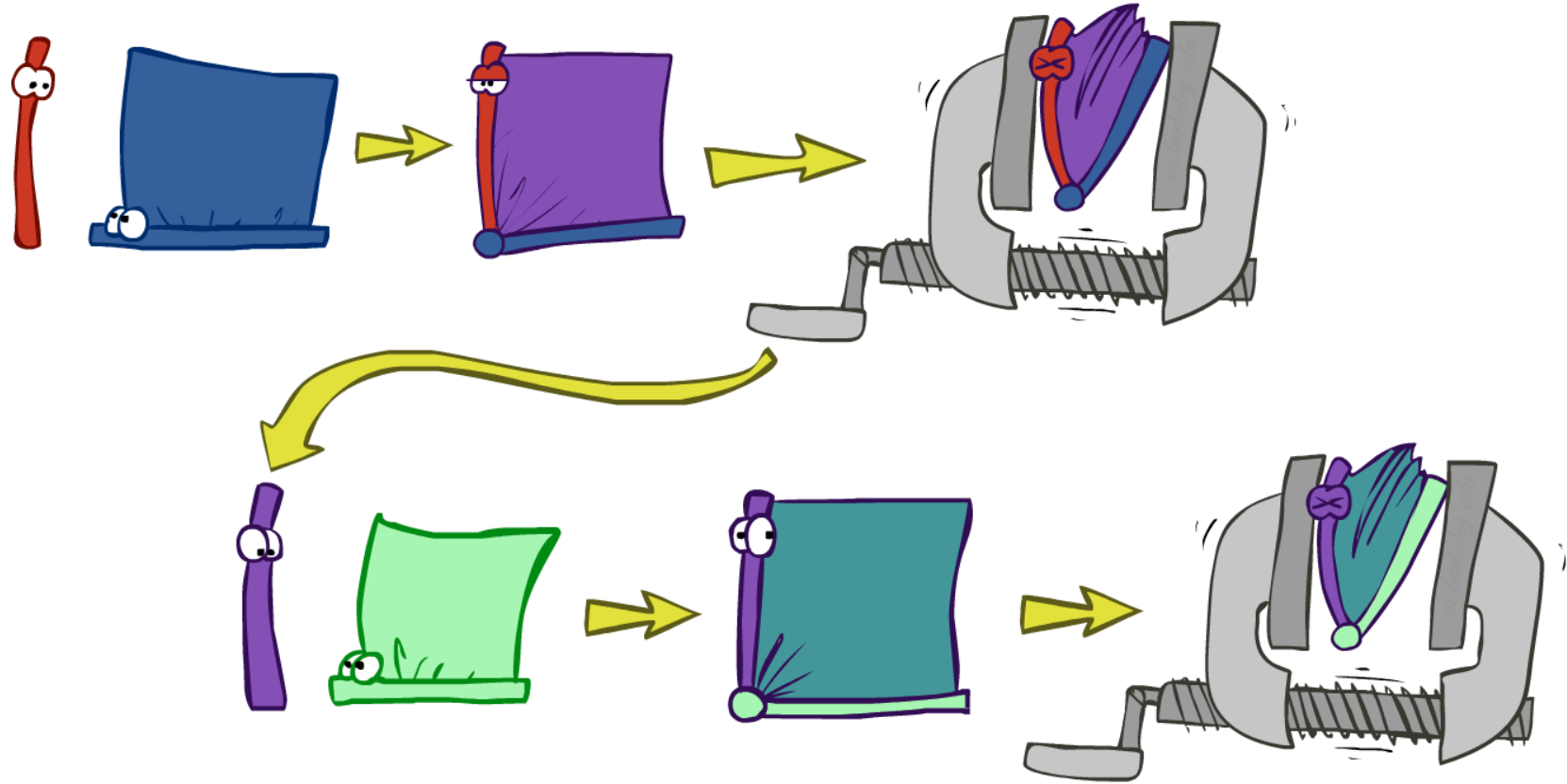


Inference by Enumeration

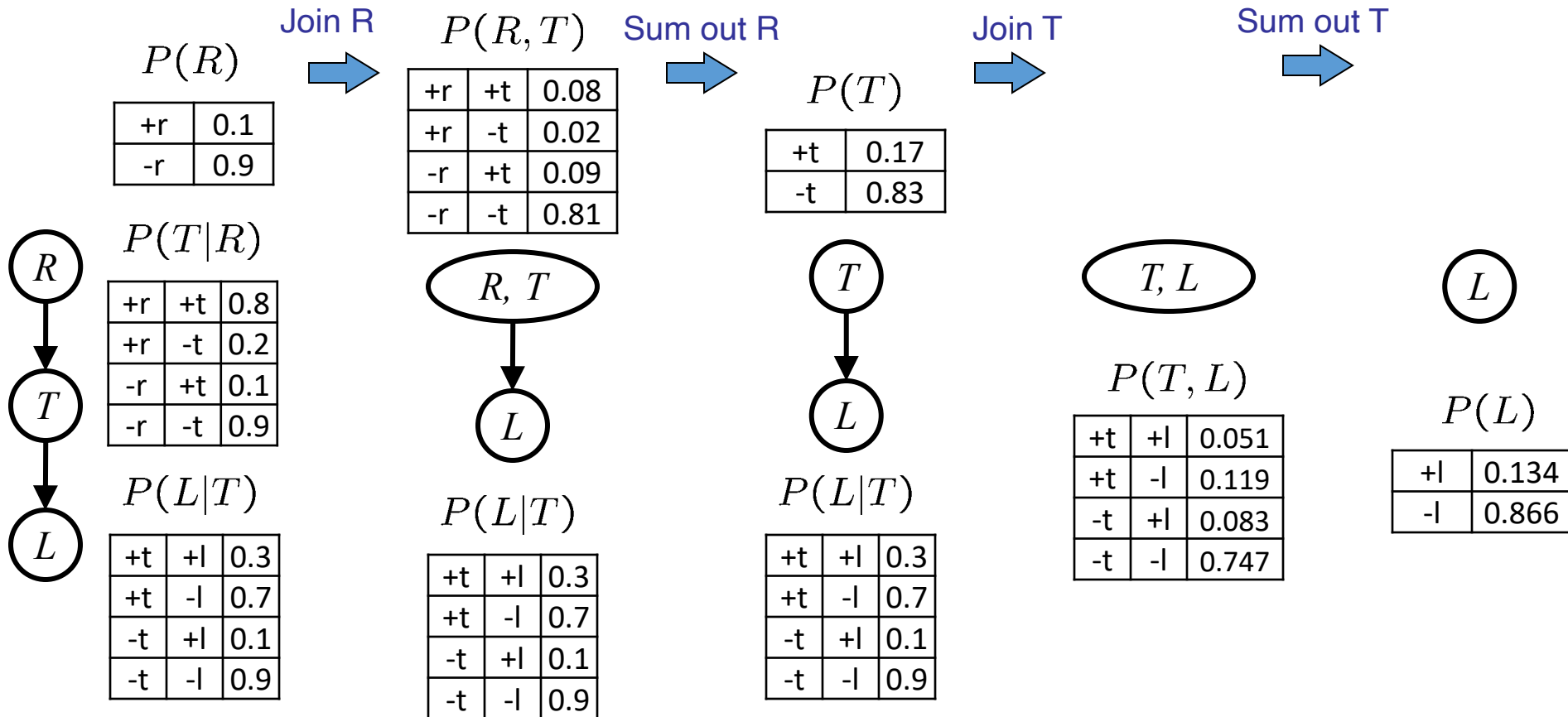
❑ Multiple joins and multiple eliminations



Marginalizing Early (= Variable Elimination)



Marginalizing Early! (aka VE)



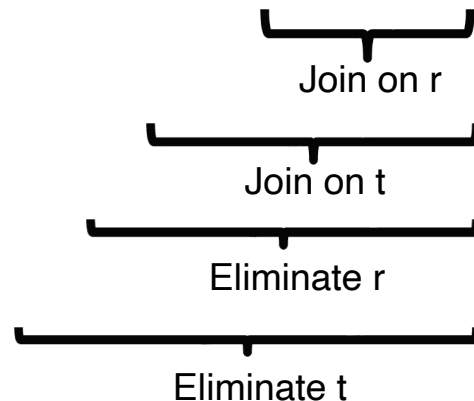
Traffic Domain

□ Inference by enumeration

□ $P(L) =$

○ $= \sum_{r,t} P(r, t, L)$

○ $= \sum_{r,t} P(L|t) P(r) P(t|r)$

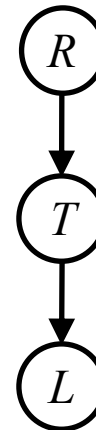
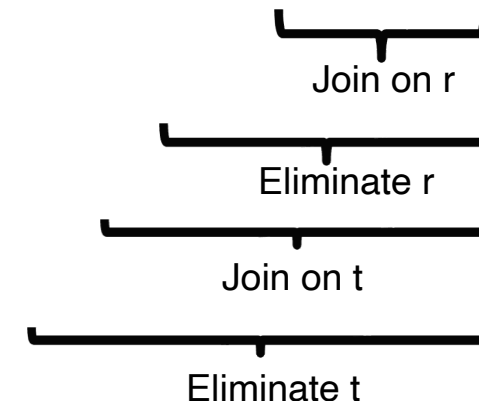


□ Variable elimination

□ $P(L) =$

○ $= \sum_{r,t} P(r, t, L)$

○ $= \sum_t P(L|t) \sum_r P(r) P(t|r)$



Evidence I

□ If evidence, start with factors that select that evidence

○ No evidence uses these initial factors:

$P(R)$		$P(T R)$			$P(L T)$		
+r	0.1	+r	+t	0.8	+t	+l	0.3
-r	0.9	+r	-t	0.2	+t	-l	0.7
		-r	+t	0.1	-t	+l	0.1
		-r	-t	0.9	-t	-l	0.9

□ Computing $P(L|r)$, the initial factors become:

$P(+r)$		$P(T +r)$			$P(L T)$		
+r	0.1	+r	+t	0.8	+t	+l	0.3
		+r	-t	0.2	+t	-l	0.7
					-t	+l	0.1
					-t	-l	0.9

□ We eliminate all variables other than query + evidence

Evidence II

❑ Result will be a selected joint of query and evidence

- E.g. for $P(L \mid r)$, we would end up with:

$P(+r, L)$			Normalize	$P(L \mid +r)$	
+r	+l	0.026		+l	0.26
+r	-l	0.074		-l	0.74

❑ To get our answer, just normalize this!

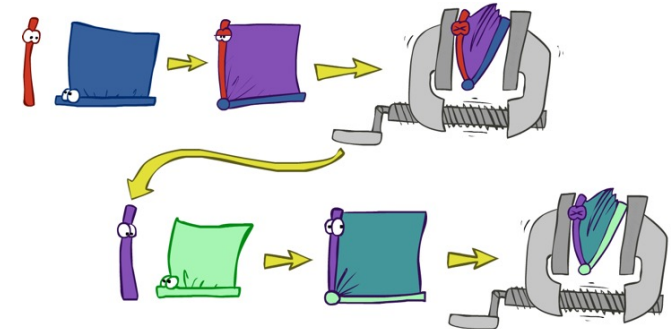
❑ That 's it!

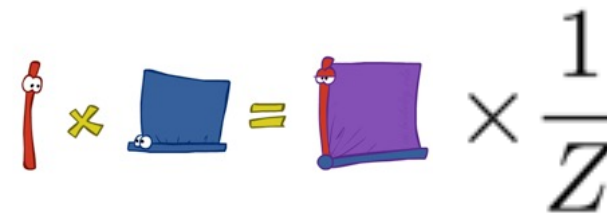
General Variable Elimination

- ❑ Query: $P(Q|e_1, \dots, e_k)$
- ❑ Start with initial factors:
 - Local CPTs (but instantiated by evidence)
- ❑ While there are still hidden variables (not Q or evidence):
 - Pick a hidden variable H
 - Join all factors mentioning H
 - Eliminate (sum out) H
- ❑ Join all remaining factors and normalize



x	P(x)
-3	0.05
-1	0.25
0	0.07
1	0.2
5	0.01

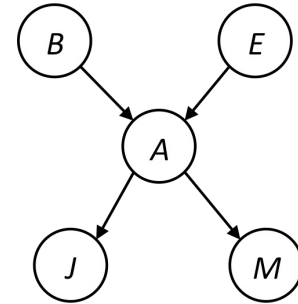



$$\text{stick figure} \times \text{blue square} = \text{purple square} \times \frac{1}{Z}$$

Example

$$P(B|j, m) \propto P(B, j, m)$$

$P(B)$	$P(E)$	$P(A B, E)$	$P(j A)$	$P(m A)$
--------	--------	-------------	----------	----------



Choose A

$$\begin{array}{l} P(A|B, E) \\ P(j|A) \\ P(m|A) \end{array} \xrightarrow{\times} P(j, m, A|B, E) \xrightarrow{\Sigma} P(j, m|B, E)$$

$P(B)$	$P(E)$	$P(j, m B, E)$
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Example

$P(B)$	$P(E)$	$P(j, m B, E)$
--------	--------	----------------

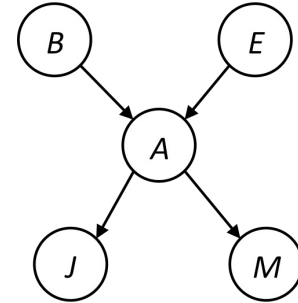
Choose E

$$\begin{array}{c} P(E) \\ P(j, m|B, E) \end{array} \xrightarrow{\times} P(j, m, E|B) \xrightarrow{\sum} P(j, m|B)$$

$P(B)$	$P(j, m B)$
--------	-------------

Finish with B

$$\begin{array}{c} P(B) \\ P(j, m|B) \end{array} \xrightarrow{\times} P(j, m, B) \xrightarrow{\text{Normalize}} P(B|j, m)$$



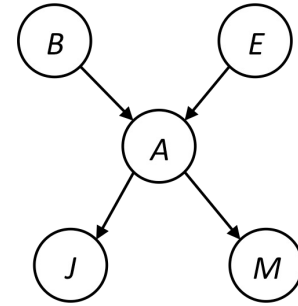
Same Example in Equations

$$P(B|j, m) \propto P(B, j, m)$$

$P(B)$	$P(E)$	$P(A B, E)$	$P(j A)$	$P(m A)$
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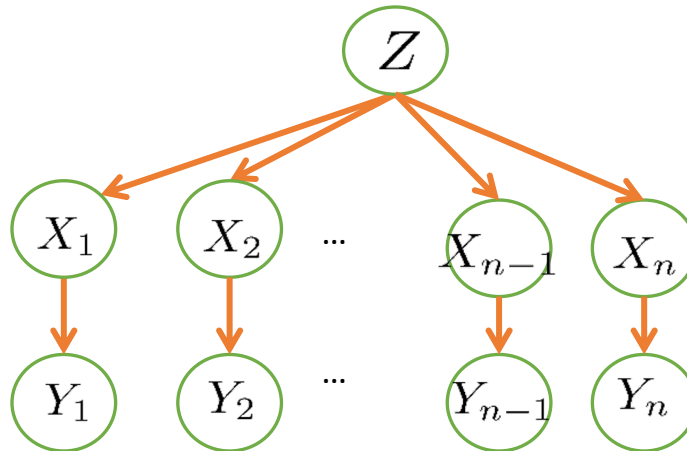
$$P(B|j, m) \propto P(B, j, m)$$

$$\begin{aligned} &= \sum P(B, j, m, e, a) \\ &= \sum_{e, a} P(B)P(e)P(a|B, e)P(j|a)P(m|a) \\ &= \sum_{e, a} P(B)P(e) \sum_a P(a|B, e)P(j|a)P(m|a) \\ &= \sum_e P(B)P(e)f_1(j, m|B, e) \\ &= P(B) \sum_e P(e)f_1(j, m|B, e) \\ &= P(B)f_2(j, m|B) \end{aligned}$$



Variable Elimination Ordering

□ For the query $P(X_n | y_1, \dots, y_n)$ work through the following two different orderings as done in previous slide: $Z, X_1, \dots, X_{n-1}$ and $X_1, \dots, X_{n-1}, Z$. What is the size of the maximum factor generated for each of the orderings?



Another Variable Elimination Example

Query: $P(X_3|Y_1 = y_1, Y_2 = y_2, Y_3 = y_3)$

Start by inserting evidence, which gives the following initial factors:

$$P(Z), P(X_1|Z), P(X_2|Z), P(X_3|Z), P(y_1|X_1), P(y_2|X_2), P(y_3|X_3)$$

Eliminate X_1 , this introduces the factor $f_1(y_1|Z) = \sum_{x_1} P(x_1|Z)P(y_1|x_1)$, and we are left with:

$$P(Z), P(X_2|Z), P(X_3|Z), P(y_2|X_2), P(y_3|X_3), f_1(y_1|Z)$$

Eliminate X_2 , this introduces the factor $f_2(y_2|Z) = \sum_{x_2} P(x_2|Z)P(y_2|x_2)$, and we are left with:

$$P(Z), P(X_3|Z), P(y_3|X_3), f_1(y_1|Z), f_2(y_2|Z)$$

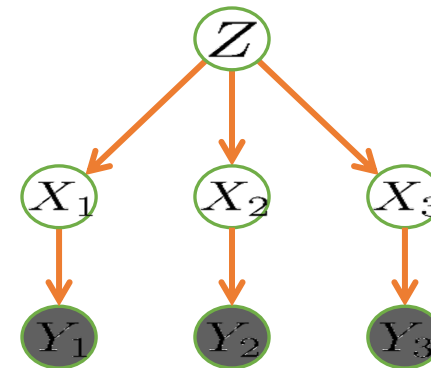
Eliminate Z , this introduces the factor $f_3(y_1, y_2, X_3) = \sum_z P(z)P(X_3|z)f_1(y_1|Z)f_2(y_2|Z)$, and we are left with:

$$P(y_3|X_3), f_3(y_1, y_2, X_3)$$

No hidden variables left. Join the remaining factors to get:

$$f_4(y_1, y_2, y_3, X_3) = P(y_3|X_3), f_3(y_1, y_2, X_3)$$

Normalizing over X_3 gives $P(X_3|y_1, y_2, y_3) = f_4(y_1, y_2, y_3, X_3) / \sum_{x_3} f_4(y_1, y_2, y_3, x_3)$



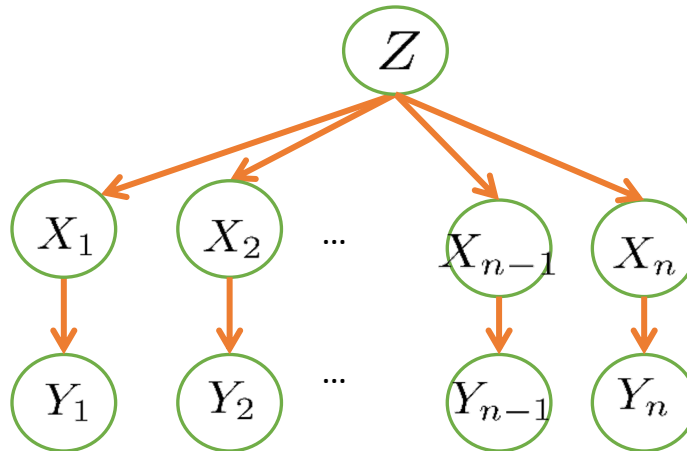
Computational complexity critically depends on the largest factor being generated in this process. Size of factor = number of entries in table. In example above (assuming binary) all factors generated are of size 2 --- as they all only have one variable (Z , Z , and X_3 respectively).

Variable Elimination Ordering

❑ For the query $P(X_n | y_1, \dots, y_n)$ work through the following two different orderings as done in previous slide: $Z, X_1, \dots, X_{n-1}$ and $X_1, \dots, X_{n-1}, Z$. What is the size of the maximum factor generated for each of the orderings?

○ Answer: 2^n versus 2 (assuming binary)

❑ In general: the ordering can greatly affect efficiency.



VE: Computational and Space Complexity

- ❑ The computational and space complexity of variable elimination is determined by the largest factor
- ❑ The elimination ordering can greatly affect the size of the largest factor.
- ❑ Does there always exist an ordering that only results in small factors?
 - No!
- ❑ Exact inference in BNs is NP-Hard

Outline

- Probability theory - basics

- Probabilistic Inference

 - Inference by enumeration

- Conditional Independence

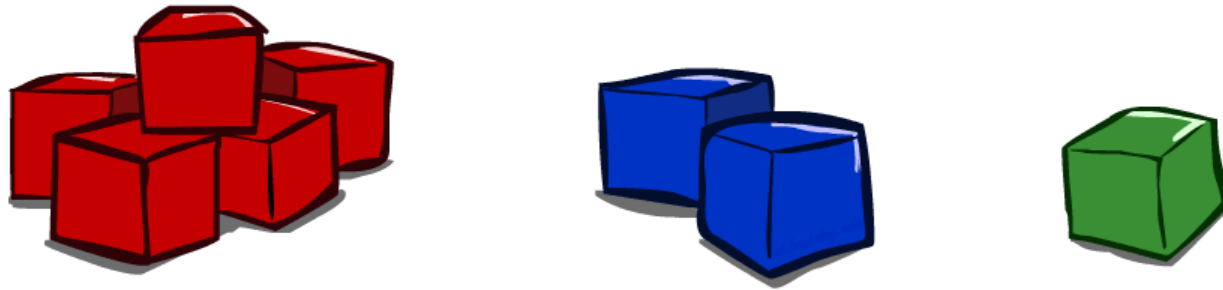
- Bayesian Networks

- Inference in Bayesian Networks

 - Exact Inference

 - Approximate Inference

Approximate Inference: Sampling



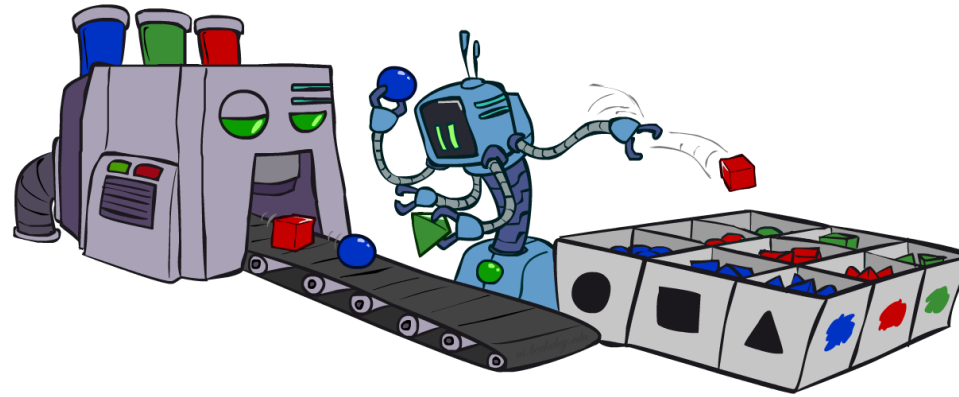
Sampling

□ Basic idea

- Draw N samples from a sampling distribution S
- Compute an approximate posterior probability
- Show this converges to the true probability P

□ Why sample?

- Inference: getting a sample is faster than computing the right answer (e.g. with variable elimination)



Sampling Basics

□ Sampling from given distribution

- Step 1: Get sample u from uniform distribution over $[0, 1)$
- E.g. `random()` in python

□ Step 2: Convert this sample u into an outcome for the given distribution by having each outcome associated with a sub-interval of $[0, 1)$ with sub-interval size equal to probability of the outcome

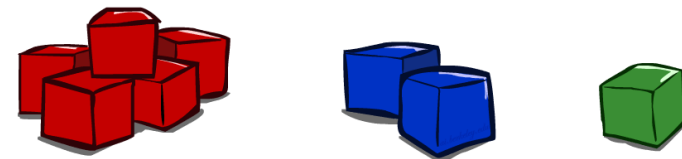
C	P(C)
red	0.6
green	0.1
blue	0.3

$$0 \leq u < 0.6, \rightarrow C = \text{red}$$

$$0.6 \leq u < 0.7, \rightarrow C = \text{green}$$

$$0.7 \leq u < 1, \rightarrow C = \text{blue}$$

- If `random()` returns $u = 0.83$, then our sample is $C = \text{blue}$
- E.g, after sampling 8 times:



Sampling in Bayes' Nets

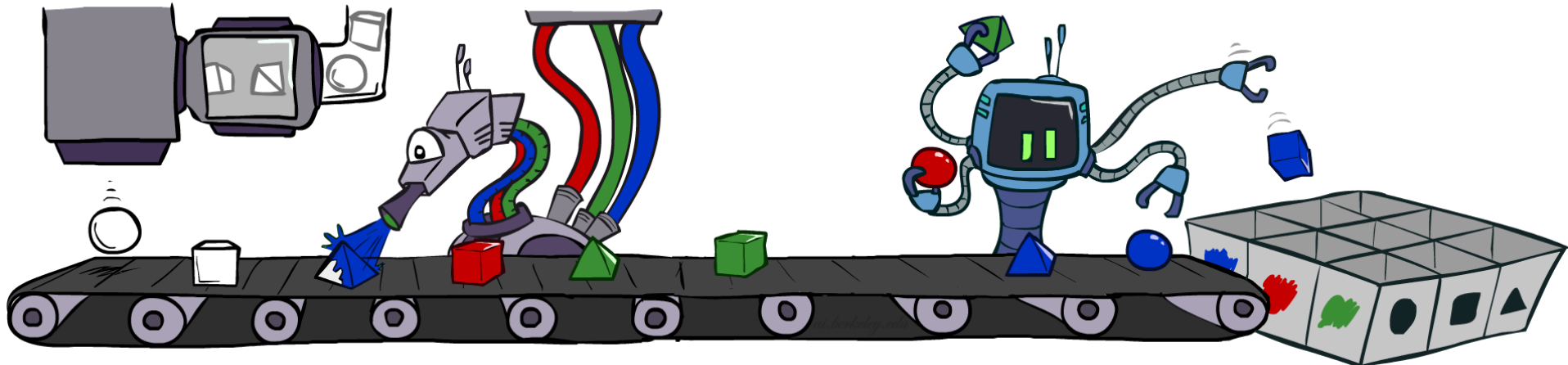
- Prior Sampling (Direct Sampling)

- Rejection Sampling

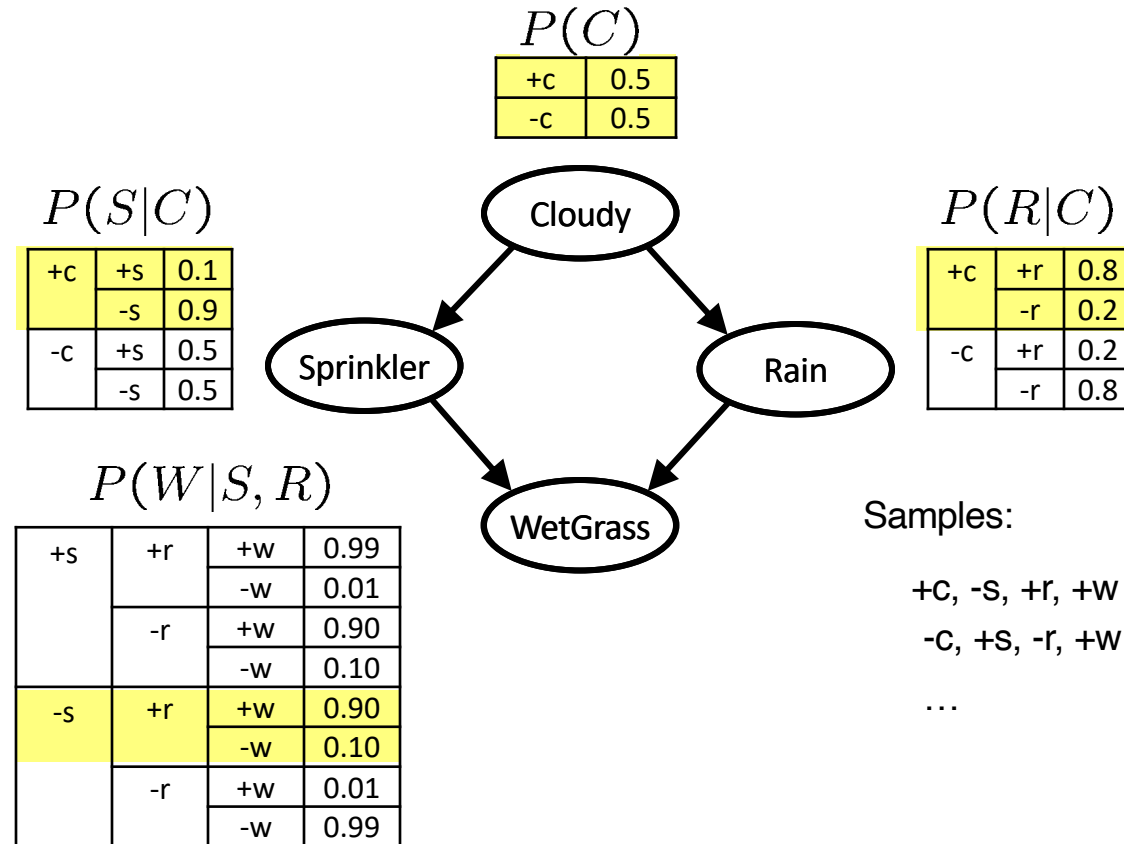
Prior (Direct) Sampling

- ❑ Ignore evidence. Sample from the joint probability.
- ❑ Do inference by counting the right samples.

- ❑ For $i = 1, 2, \dots, n$
 - Sample x_i from $P(X_i \mid \text{Parents}(X_i))$
- ❑ Return $(x_1, x_2, \dots, x_n)$



Prior Sampling



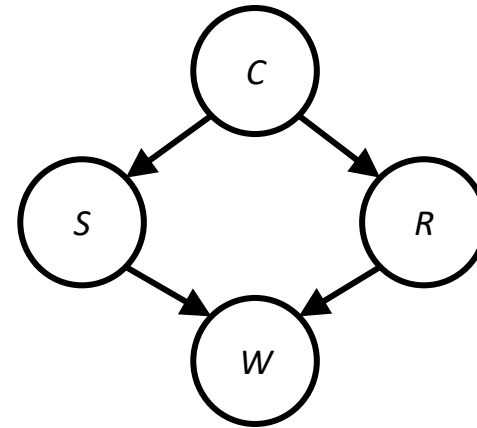
Example

□ We'll get a bunch of samples from the BN:

+C, -S, +r, +W
+C, +S, +r, +W
-C, +S, +r, -W
+C, -S, +r, +W
-C, -S, -r, +W

□ If we want to know $P(W)$

- We have counts $\langle +w:4, -w:1 \rangle$
- Normalize to get $P(W) = \langle +w:0.8, -w:0.2 \rangle$
- This will get closer to the true distribution with more samples
- Can estimate anything else, too
- What about $P(C \mid +w)$? $P(C \mid +r, +w)$? $P(C \mid -r, -w)$?



Prior Sampling Analysis

- This process generates samples with probability:

$$S_{PS}(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{Parents}(x_i)) = P(x_1, \dots, x_n)$$

- ...i.e. the BN's joint probability

- Let the number of samples of an event be $N_{PS}(x_1, \dots, x_n)$

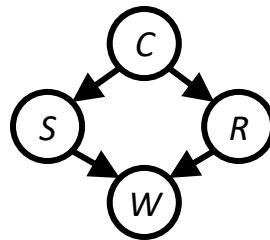
- Then $\lim_{N \rightarrow \infty} \hat{P}(x_1, \dots, x_n) = \lim_{N \rightarrow \infty} N_{PS}(x_1, \dots, x_n) / N$
 $= S_{PS}(x_1, \dots, x_n) = P(x_1, \dots, x_n)$

- I.e., the sampling procedure is **consistent**

Rejection Sampling

□ Let's say we want $P(C | +s)$

- Tally C outcomes, but ignore (reject) samples which don't have $S = +s$
- This is called rejection sampling
- It is also consistent for conditional probabilities (i.e., correct in the limit)



+C, -S, +r, +W
+C, +S, +r, +W
-C, +S, +r, -W
+C, -S, +r, +W
-C, -S, -r, +W

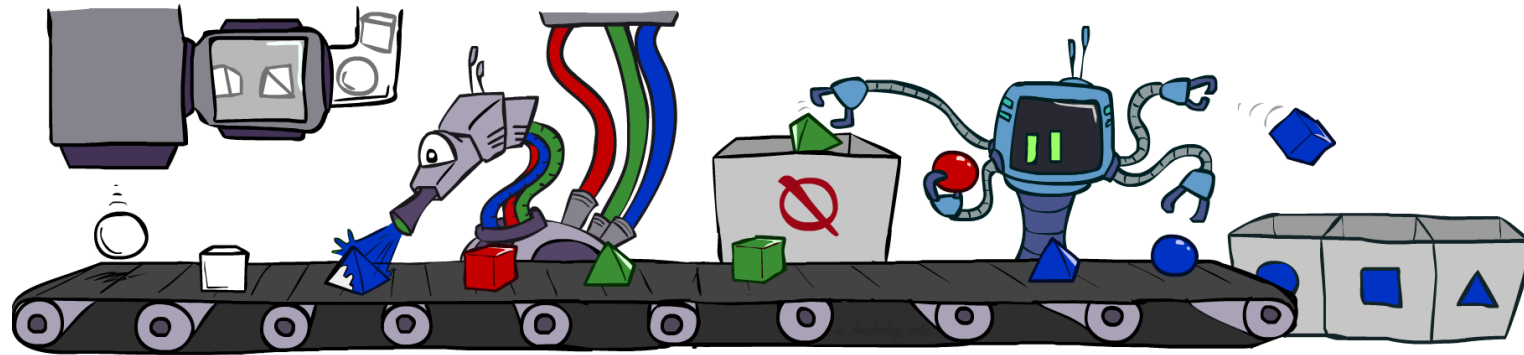
Rejection Sampling

❑ Evidence instantiation

❑ For $i = 1, 2, \dots, n$

- Sample x_i from $P(X_i \mid \text{Parents}(X_i))$
- If x_i not consistent with evidence
- Reject: Return, and no sample is generated in this cycle

❑ Return $(x_1, \dots, x_n)$



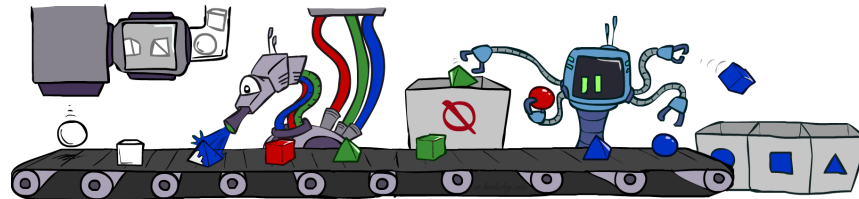
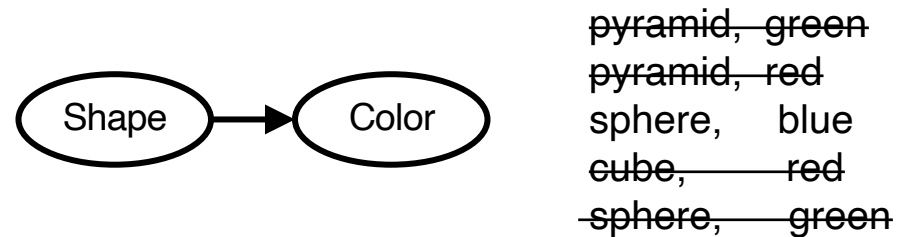
Rejection Sampling Analysis

- Let $\hat{P}(X|e)$ be the estimated distribution that the algorithm returns.
- By definition: $\hat{P}(X|e) = \alpha N_{PS}(X, e) = \frac{N_{PS}(X, e)}{N_{PS}(e)}$
- We already know that
 - $N_{PS}(x_1, \dots, x_n)/N \approx P(x_1, \dots, x_n)$ and
 - $N_{PS}(x_1, \dots, x_m)/N \approx P(x_1, \dots, x_m)$ for $m < n$
- Thus $\hat{P}(X|e) \approx P(X|e)/P(e) = P(X|e)$
- Sampling procedure is consistent

Rejection Sampling

❑ Drawback?

- If evidence is unlikely, rejects lots of samples
- Evidence not exploited as you sample
- Consider $P(\text{Shape}|\text{blue})$



Outline

- Probability theory - basics

- Probabilistic Inference

 - Inference by enumeration

- Conditional Independence

- Bayesian Networks

- Inference in Bayesian Networks

 - Exact Inference

 - Approximate Inference